Description

Image





Caption

1. Stainless steel is commonly used where corrosion must be avoided, such as in household scissors. (c) ANSYS, Inc. 2. The sunburst-patterned stainless steel spire of the Chrysler building is an iconic part of the New York skyline. 3. Stainless steel pipes used in food manufacturing. Images 2 and 3 used under license from Shutterstock.com

The material

Stainless steels are alloys of iron with chromium, nickel, and - often - four or five other elements. The alloying transmutes plain carbon steel that rusts and is prone to brittleness below room temperature into a material that does neither. Indeed, most stainless steels resist corrosion in most normal environments, and they remain ductile to the lowest of temperatures.

Compositional summary

Fe/<0.25C/16 - 30Cr/3.5 - 37Ni/<10Mn + Si,P,S (+N for 200 series)

General properties

Density		7.61e3	-	7.87e3	kg/m^3
Price	*	5.32	-	5.81	GBP/kg
Date first used		1915			

Mechanical properties

Young's modulus		190	-	210	GPa
Shear modulus		74	-	82	GPa
Bulk modulus		140	-	160	GPa
Poisson's ratio		0.27	-	0.28	
Yield strength (elastic limit)		257	-	1.14e3	MPa
Tensile strength		515	-	1.3e3	MPa
Compressive strength	*	252	-	1.2e3	MPa

Elongation		10	-	49	% strain
Hardness - Vickers		170	-	438	HV
Fatigue strength at 10^7 cycles	*	256	-	542	MPa
Fracture toughness	*	57	-	137	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	3.1e-4	-	0.0012	

Thermal properties

Melting point		1.4e3	-	1.49e3	°C
Maximum service temperature		640	-	747	°C
Minimum service temperature	*	-150	-	-73.2	°C
Thermal conductor or insulator?		Poor conductor			
Thermal conductivity		14	-	24.9	W/m.°C
Specific heat capacity		450	-	510	J/kg.°C
Thermal expansion coefficient		10.8	-	16.5	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Poor conductor			
Electrical resistivity		64	-	87	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		3	-	4	
Formability		2	-	3	
Machinability		2	-	3	
Weldability		5			
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Excellent
Nitric acid (70%)	Limited use
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Acceptable
Benzene	Acceptable
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Acceptable
Toluene	Excellent
Turpentine	Acceptable
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Acceptable
Formaldehyde (40%)	Acceptable

Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Excellent
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Excellent
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.23e9			tonne/yr
Reserves, principal component	1.59e11			tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 5.73	-	6.32	kg/kg
Embodied energy (virgin grade production)	* 66.4	-	73.2	MJ/kg
Water usage	* 129	-	142	l/kg

Material processing: energy

Metal casting, embodied energy	* 47.9	-	54.5	MJ/kg
Metal extrusion, hot, embodied energy	* 31.7	-	70.9	MJ/kg
Metal rolling and forging, embodied energy	* 16.4	-	36.2	MJ/kg
Wire drawing, embodied energy	* 4.38	-	4.83	MJ/kg

Metal powder forming, embodied energy	* 66.8	-	78.8	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 9.63e3	-	1.17e4	MJ/kg
Machining, coarse, embodied energy	* 19.2	-	26.6	MJ/kg
Machining, fine, embodied energy	* 28.5	-	47.6	MJ/kg
Machining, grinding, embodied energy	* 37.8	-	68.7	MJ/kg
Machining, laser, CO2, embodied energy	* 1.15	-	1.39	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.45	-	3.95	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 2.07	-	4.65	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 1.07	-	2.37	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.286	-	0.317	kg/kg
Metal powder forming, climate change (CO2-eq)	* 5.45	-	6.43	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 793	-	960	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.26	-	1.74	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.86	-	3.12	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.47	-	4.5	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0944	-	0.114	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.55	-	1.71	kg/kg
Embodied energy, recycling	* 20.6	-	22.7	MJ/kg
Recycle fraction in current supply	34			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Stainless steels are FDA approved -- indeed, they are so inert that they can be implanted in the body, and are widely used in food processing equipment. All can be recycled.

Supporting information

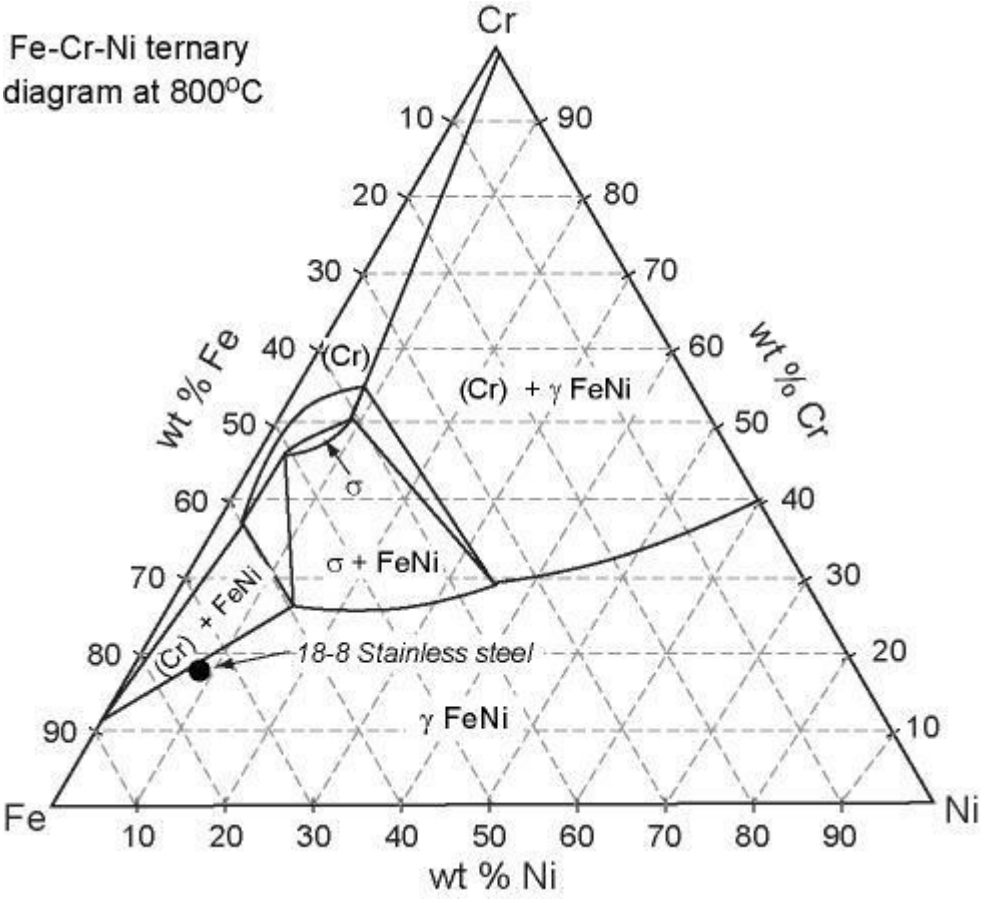
Design guidelines

Stainless steel must be used efficiently to justify its higher costs, exploiting its high strength and corrosion resistance. Economic design uses thin, rolled gauge, simple sections, concealed welds to eliminate refinishing, and grades that are suitable to manufacturing (such as free machining grades when machining is necessary). Surface finish can be controlled by rolling, polishing or blasting. Stainless steels are selected, first, for their corrosion resistance, second, for their strength and third, for their ease of fabrication. Most stainless steels are difficult to bend, draw and cut, requiring slow cutting speeds and special tool geometry. They are available in sheet, strip, plate, bar, wire, tubing and pipe, and can be readily soldered and braised. Welding stainless steel is possible but the filler metal must be selected to ensure an equivalent composition to maintain corrosion resistance. The 300 series are the most weldable; the 400 series are less weldable.

Technical notes

Stainless steels are classified into four categories: the 200and 300 series austenitic (Fe-Cr-Ni-Mn) alloys, the 400 series ferritic (Fe-Cr) alloys, the martensitic (Fe-Cr-C) alloys that also form part of the 400 series, and precipitation hardening or PH (Fe-Cr-Ni-Cu-Nb) alloys with designations starting with S. Typical of the austenitic grades of stainless steel is the grade 304: 74% iron, 18% chromium and 8 % nickel. Here the chromium protects by creating a protective Cr2O3 film on all exposed surfaces, and the nickel stabilizes face-centered cubic austenite, giving ductility and strength both at high and low temperatures; they are non-magnetic (a way of identifying them). The combination of austenitic and ferritic structures (the duplex stainless steels) provide considerably slower growth of stress-induced cracks, they can be hot-rolled or cast and are often heat treated as well. Austenitic stainless steel with high molybdenum content and copper has excellent resistance to pitting and corrosion. High nitrogen content austenitic stainless steel gives higher strength. Superferrites (over 30% chromium) are very resistant to corrosion, even in water containing chlorine.

Phase diagram



Phase diagram description

Most stainless steels are alloys of iron (Fe) with chromium (Cr) and nickel (Ni). This is the ternary phase diagram, at a temperature of 800 C, for those three elements. The position of AISI 302 stainless steel (Fe-18%Cr-8%Ni) is shown.

Typical uses

Railway cars, trucks, trailers, food-processing equipment, sinks, stoves, cooking utensils, cutlery, flatware, scissors and knives, architectural metalwork, laundry equipment, chemical-processing equipment, jet-engine parts, surgical tools, furnace and boiler components, oil-burner parts, petroleum-processing equipment, dairy equipment, heat-treating equipment, automotive trim. Structural uses in corrosive environments, e.g. nuclear plants, ships, offshore oil installations, underwater cables and pipes.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.